Topic 4: Python-Based Transfer Learning for 3D Printed Object Defect Detection

Title:

Transfer Learning-Based Defect Detection in 3D-Printed Objects

Dataset:

This topic uses image datasets of 3D-printed objects with defects. Python can be used with pretrained CNN models such as ResNet, VGG, MobileNet, EfficientNet, YOLOv5, and YOLOv8 for detecting defects in printed parts.

Dataset URL: https://github.com/mahsan2/AMdataset

GitHub project repository: https://github.com/mahsan2/AM3d

Latest Paper:

- Defect analysis of 3D printed object using transfer learning approaches: https://doi.org/10.1016/j.eswa.2024.124293

使用 Python 迁移学习检测 3D 打印工件缺陷

项目名称

基于迁移学习的 3D 打印工件缺陷检测

数据集

本选题使用含有缺陷的 3D 打印工件图像数据集。可以使用 Python 搭配 ResNet、VGG、MobileNet、EfficientNet、YOLOv5 和 YOLOv8 等预训练 CNN 模型，检测打印工件中的缺陷。

数据集链接

https://github.com/mahsan2/AMdataset

GitHub 项目仓库

https://github.com/mahsan2/AM3d

最新论文

《使用迁移学习方法分析 3D 打印工件缺陷》

https://doi.org/10.1016/j.eswa.2024.124293